

RESEARCH ARTICLE

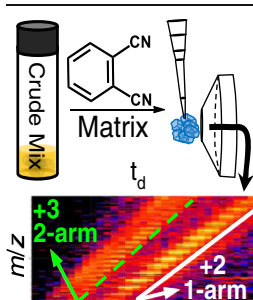
Matrix-Assisted Ionization-Ion Mobility Spectrometry-Mass Spectrometry: Selective Analysis of a Europium–PEG Complex in a Crude Mixture

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Abstract. The analytical utility of a new and simple to use ionization method, matrix-assisted ionization (MAI), coupled with ion mobility spectrometry (IMS) and mass spectrometry (MS) is used to characterize a 2-armed europium(III)-containing poly(ethylene glycol) (Eu-PEG) complex directly from a crude sample. MAI was used with the matrix 1,2-dicyanobenzene, which affords low chemical background relative to matrix-assisted laser desorption/ionization (MALDI) and electrospray ionization (ESI). MAI provides high ion abundance of desired products in comparison to ESI and MALDI. Inductively coupled plasma-MS measurements were used to estimate a maximum of 10% of the crude sample by mass was the 2-arm Eu-PEG complex, supporting evidence of selective ionization of Eu-PEG complexes using the new MAI

matrix, 1,2-dicyanobenzene. Multiply charged ions formed in MAI enhance the IMS gas-phase separation, especially relative to the singly charged ions observed with MALDI. Individual components are cleanly separated and readily identified, allowing characterization of the 2-arm Eu-PEG conjugate from a mixture of the 1-arm Eu-PEG complex and unreacted starting materials. Size-exclusion chromatography, liquid chromatography at critical conditions, MALDI-MS, ESI-MS, and ESI-IMS-MS had difficulties with this analysis, or failed.

Keywords: Selective ionization matrix-assisted ionization ion mobility spectrometry mass spectrometry, Europium, Poly(ethylene glycol), Size-exclusion chromatography, Liquid chromatography at critical conditions, Electrospray ionization, Matrix-assisted laser desorption/ionization, Inductively coupled plasma-mass spectrometry

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Introduction

Recently, a new ionization process was discovered that produces ions from volatile and nonvolatile compounds, ranging from drugs to proteins, similar in charge state and abundance to electrospray ionization (ESI) [1–3]. Matrix-assisted ionization (MAI) operates directly from the solid

state without high voltages or a laser, overcoming issues including solvent- and salt-spray compatibility common to liquid chromatography (LC)-ESI-mass spectrometry (MS) that result from the need to apply high voltages [4, 5], particularly with synthetic polymers [6, 7]. The initial applications of MAI, primarily peptides, proteins, and drugs, were enabled using 3-nitrobenzonitrile (3-NBN) as a matrix [1–3, 8, 9], which showed a suppression of salt adduction. However, for the same reasons that 3-NBN produces low background ions, it ionizes synthetic polymers, typically ionized through salt adduction, with poor efficiency [9, 10]. The discovery of over 40 MAI matrices, including 1,2-dicyanobenzene (1,2-DCB) and 2-bromo-2-nitro-1,3-propanediol (bronopol) has

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enhanced the applicability of MAI with synthetic polymers [10].

Due to the inherent polydispersity within polymeric samples, methods that produce multiply charged ions generate complex mass spectra with overlapping polymer distributions [11]. To address issues of sample complexity, multiple methods are often used to characterize polymeric materials. Ion mobility spectrometry (IMS), a post-ionization gas-phase separation method that differentiates ions by charge, size, and shape has been coupled to MS to provide detailed information of complex polymeric samples in the form of drift time (t_d) versus mass-to-charge ratio (m/z) measurements. The addition of the t_d dimension provided by IMS enhances polymer analysis by affording a method of deconvoluting overlapping charge state distributions within polymer distributions [12, 13] as well as polymer complexity associated with structural composition [12–19]. The generation of multiply charged ions can increase IMS separation and potentially enhance the dynamic range of the experiment [12].

Matrix-assisted laser desorption/ionization (MALDI)-time-of-flight (TOF)-MS is traditionally used to analyze individual chain lengths of polymers [20–23] while providing high sensitivity and, in principle, an unlimited mass range [24, 25]. MALDI is advantageous for polymer characterization primarily because singly charged ions are dominant [26, 27], producing straightforward mass spectra with fewer overlapping polymer distributions. Electrospray ionization (ESI) produces more complex mass spectra because it produces higher charge states [28, 29], but is advantageous in being a softer ionization method and thus is applicable to characterization of fragile complexes [30, 31]. This includes identification of monomer units and end group analysis, which are usually determined by observed m/z [32]. Therefore, for more complex polymer architectures and blends, a combination of analytical methods such as the coupling of size-exclusion chromatography (SEC) or liquid chromatography at critical conditions (LCCC) with MALDI-MS or ESI-MS are often used [33–37]. SEC provides average molecular weight information through the use of standard calibrants. However, SEC does not provide information regarding structural differences that can contribute to similarities in the molecular masses of different species within a mixture [38]. Therefore, methods complementary to SEC are often necessary [39–41]. For example, LCCC separates polymers based on structural differences, including functionality type distribution, composition distribution of copolymers, or differences in microstructure or topology independent of the molecular mass [32, 42, 43]. Approaches hyphenating LC to inductively coupled plasma (ICP)-MS have been used to quantify metal complexes within other crude, biological samples [44–47], including synthetic polymers [48].

The prevalent use of magnetic resonance imaging (MRI) as a tool in clinical medicine and biomedical research is a result of the useful features of the technique, including its noninvasive nature [49–52]. Recently, a europium based poly(ethylene

glycol) (PEG) 2-arm conjugate (Eu-PEG) was synthesized to study fundamental properties relating to contrast agent performance [52, 53] but the analyses of Eu-PEG complex (net charge +1) was difficult. Here, we use this crude sample exemplifying the intricate nature of standard polymer characterization methods and demonstrate the unique ionization and separation abilities of MAI-IMS-MS for Eu-PEG complexes directly from the crude sample.

Material and Methods

Materials and Synthesis of Europium(III)-Containing Polymer Conjugates

Matrix compounds trans-2-[3-(4-*tert*-butylphenyl)-2-methyl-2-propenylidene]malononitrile (DCTB), 3-NBN, bronopol, and 1,2-DCB were purchased from Sigma Aldrich (St. Louis, MO, USA). Acetonitrile (ACN), methanol (MeOH), and high performance LC-grade water were obtained from Fisher Scientific (Pittsburgh, PA, USA). Synthesis of the 2-arm (~2300 Da) Eu-PEG conjugate was carried out as previously described [53].

Chromatography: Methods and Instrumentation

Aqueous SEC was carried out on a Shimadzu HPLC system (Shimadzu Corporation, Kyoto, Japan) equipped with three aquagel-OH columns in series (VARIAN PLaquagel-OH-mixed, 8 $\mu\text{m} \times 300$ mm) or a Bio-Sil SEC 250 column (BIO-RAD gel filtration HPLC column, 5 $\mu\text{m} \times 300$ mm), with fluorescence ($\lambda_{\text{ex}}=396$ nm and $\lambda_{\text{em}}=593$ nm for Eu(III)-containing conjugates), photodiode array (followed by monitoring absorbance at 210 nm), and refractive index detectors. An isocratic method of 100% H₂O with a flow rate of 1 mL min⁻¹ was used with the aquagel-OH columns, and flow rates of 0.7 or 0.5 mL min⁻¹ were used with the Bio-Sil SEC 250 column. SEC was also performed using commercially available Sephadex G-25 and Sephadex G-10 with H₂O as the mobile phase under gravity flow, where G-25 and G-10 correspond to fractionation range of the dextrin gel type, with larger G values corresponding to larger molecular weight fractionation ranges.

LCCC was performed on an Agilent 1100/1200 Series HPLC equipped with an evaporative light scattering detector (ELSD) (PL-ELS 1000; Polymer Labs, now Agilent Technologies, Santa Clara (CA), USA). The ELSD nebulizer temperature was set to 90°C and the evaporator temperature to 110°C. The solvent composition was 47:53 ACN/H₂O (v:v) as determined to be the critical conditions for PEG. The crude samples were dissolved as 1 mg mL⁻¹, and 10 μL of the sample was injected at a flow rate of 0.5 mL min⁻¹. The column used was an octadecyl silica column with a pore size of 120 Å and particle size of 5 μm with a 250 $\times$ 4.6 mm inner diameter.

Mass Spectrometry: Methods and Instrumentation

MALDI-TOF-MS was performed on a Bruker Autoflex III TOF/TOF (Bruker Daltonic, Bremen, Germany) mass

spectrometer. DCTB was used as the MALDI matrix and was prepared as 10 mg mL^{-1} in ACN. The Eu-PEG mixture was prepared as 1 mg mL^{-1} in 47:53 ACN/H₂O. The matrix was premixed with the analyte and spotted onto a target plate and allowed to dry. An arbitrary laser fluence of 15% was used to desorb the matrix:analyte from the plate. Data was processed using FlexAnalysis v3.3 (Bruker Daltonic).

Intermediate pressure MALDI, atmospheric pressure ESI, and MAI were performed on two Waters SYNAPT G2 quadrupole-traveling wave IMS-TOF (Q-IMS-TOF) mass spectrometers (Waters Corporation, Manchester, UK). Mass spectral data was processed using MassLynx v4.1 (Waters Corporation, Milford, MA), and two-dimensional (2-D) plots were processed using DriftScope v2.3 (Waters Corporation, Manchester, UK). No cationization agent was used for any experiment. A logarithmic setting was used while processing data using DriftScope to visualize the relative ion abundance. The logarithm increases low abundant signals relative to the most abundant signal, allowing visualization of minor components present in the crude sample. This operation distorts the relativity of the ion signals so that the 2-D plot is not linearly representative of ion abundances. To this end, each species identified on the 2-D plot was individually extracted to determine the relative ion abundance. The 2-D plots use a false color ('Hot Metal') plot to display relative ion abundance, ranging from blue as the least abundant through red to yellow as the most abundant.

For MALDI-IMS-MS, the matrix used was α -cyano-4-hydroxycinnamic acid with no cationization agent. In MALDI, arbitrary laser fluences of 150 to 250 were used. IMS-MS was performed with N₂ as the ion mobility gas with the following ion mobility settings: 40 V wave height and 650 m s^{-1} wave velocity. The IMS DC settings were as follows: 25.0, 35.0, -5.0, 3, and 0 for the IMS DC entrance, helium cell DC, helium exit, IMS bias, and IMS DC exit, respectively.

For ESI-IMS-MS analysis, the dissolved Eu-PEG crude sample (0.10 mg mL^{-1} , 47:53 ACN/H₂O) was infused at a flow rate of $10 \mu\text{L min}^{-1}$ with a capillary voltage of 3.3 kV. The sampling cone and extraction cone voltages were set to 50 and 4, respectively. The source block temperature was 50°C , and the desolvation temperature was 250°C .

MAI-IMS-MS was performed on the atmospheric pressure Z-Spray source of the Waters SYNAPT G2 with the ESI housing removed as previously described [3]. Three MAI matrices (1,2-DCB, 3-NBN, and bronopol) were prepared as solutions of 5 mg in 50 μL of 75:25 ACN/H₂O, ACN, and 75:25 ACN/H₂O, respectively. The matrices were premixed with the crude sample of the Eu-PEG conjugate (1.0 mg mL^{-1}). Matrix:analyte was mixed 1:1 (v/v) and drawn into a pipet tip in 1 μL aliquots and allowed to crystallize at the end of the tip. The samples were introduced directly into the vacuum of the mass spectrometer by holding the pipet tip close to the commercial skimmer cone inlet aperture and allowing the matrix:analyte to be drawn in by the pressure differential between atmospheric pressure and the vacuum of the mass spectrometer. The source block temperature was set to 50°C , and

the voltages of the sampling and extraction cones were set to 50 and 4, respectively.

Determination of Europium Mass % in a Crude Sample Using Inductively Coupled Plasma

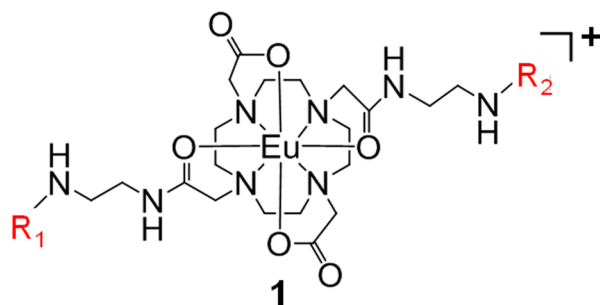
ICP-MS measurements were taken using an Agilent 7700x ICP mass spectrometer. All samples were diluted with 2% HNO₃, which was also used as the blank sample during calibration. The calibration curve was created by measuring counts per second of Eu-153 for a 1–500 ppm concentration range (diluted from Alfa Aesar, Ward Hill (MA), USA Specpure AAS standard solution, Eu₂O₃ in 5% HNO₃, $1000 \mu\text{g mL}^{-1}$). The crude Eu-PEG sample was then diluted to fit within the concentration range of the calibration curve.

Results

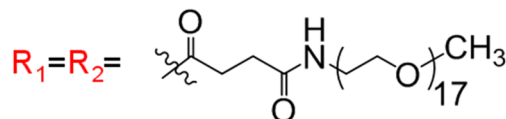
Traditional LC-MS Polymer Characterization Methods

Possible structures of products and relevant starting materials are provided in Scheme 1 and Supplementary Scheme S1. Purification of crude samples containing the 2-arm (Scheme 1a) and 1-arm (Scheme 1b) Eu-PEG conjugates was initially attempted using SEC with H₂O as the mobile phase. Pure fractions of the 2- and 1-arm products could not be

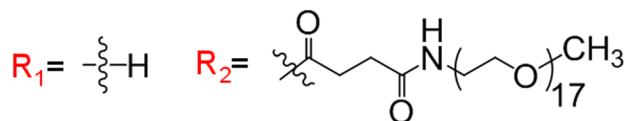
Europium-containing Conjugates



a. 2-arm Product



b. 1-arm Product



Scheme 1. Predicted structures of the (a) 1-arm (MW ~1500) and (b) 2-arm (MW ~2300) Eu-PEG products synthesized from a europium(III)-containing core **1** (Supplementary Scheme S1). The number of PEG monomers represents the average value of the commercially available starting material. Structures of the starting materials are included in Supplementary Scheme S1

obtained. The crude sample was then analyzed using a Bruker MALDI-TOF mass spectrometer to investigate the sample composition. An intense, singly charged polydisperse distribution was observed with the most abundant ion at m/z 884 (Supplementary Figure S1) and the expected ethylene oxide repeat unit of 44, indicating the presence of PEG within the sample. A comparison to the succinimidyl ester methyl ether PEG starting material (Supplementary Scheme S1a) provided identical mass spectra (Supplementary Figure S1), suggesting that the unreacted PEG starting material was present in high abundance in the crude sample. Insights from SEC and MALDI pointed to the need of more sophisticated separation methods to characterize this sample. Therefore, off-line LCCC separation optimized for PEG was used in which the fractions were characterized by MALDI-TOF-MS. Unfortunately, off-line LCCC coupled to MALDI-TOF-MS was also unable to separate the 2- and 1-arm Eu-PEG product from the crude mixture despite using critical conditions of separation (Supplementary Figures S2 and S3). The inability to isolate the 2-arm product was unfortunate because it cannot be used for fundamental studies relevant to contrast agents for MRI. Therefore, an IMS-MS study was performed to observe the 2-arm complex (molecular weight 2362.2 Da, with a net charge of +1) and learn about the synthesis through the composition of the crude mixture.

Solvent-Free IMS-MS Gas-Phase Separation for Oligomer Characterization

MALDI ionization using the SYNAPT G2 equipped with IMS afforded singly charged ions Figure 1.I.a, most visibly the PEG starting material (represented with a dashed grey line, Figure 1.I.b), along with an intense chemical background above m/z 1200 (represented by the broad, red distribution in the 2-D plot, Figure 1.I.b). The strong chemical background generated above m/z 1200, where singly charged Eu-PEG species are expected, suppressed identification. However, the 2-D plot generated by IMS-MS showed the presence of other species that were not the PEG starting material, though poorly separated in the t_d dimension (Figure 1.I.b). Deconstruction of the 2-D plot into mass spectra (Supplementary Figure S4) through selection of corresponding t_d and m/z regions (hereon referred to as ‘extraction’) made the 2- and 1-arm Eu-PEG conjugates observable, with oligomer distributions centered about m/z 2362.3 and m/z 1500.9, indicated with a green and blue line, respectively (Figure 1.I.b). Both complexes detected were singly charged. The unique isotopic distribution afforded by europium assisted in identifying the europium-containing complexes. The 2- and 1-arm Eu-PEG products are separated from each other in the m/z dimension (Supplementary Figure S4).

Multiply charged protonated, sodiated, and potassiated ions were produced using ESI from the crude sample (Figure 1.II.a), filling the IMS-MS space more efficiently than with MALDI ionization (Figure 1.I.b), demonstrating improved gas-phase separation relative to MALDI-IMS-MS (Figure 1.I.b). Ions corresponding to singly charged PEG starting material, doubly charged 1-arm, and triply charged 2-arm products are all isobars overlapping in the m/z dimension (Figure 1.II.b). Interpretation

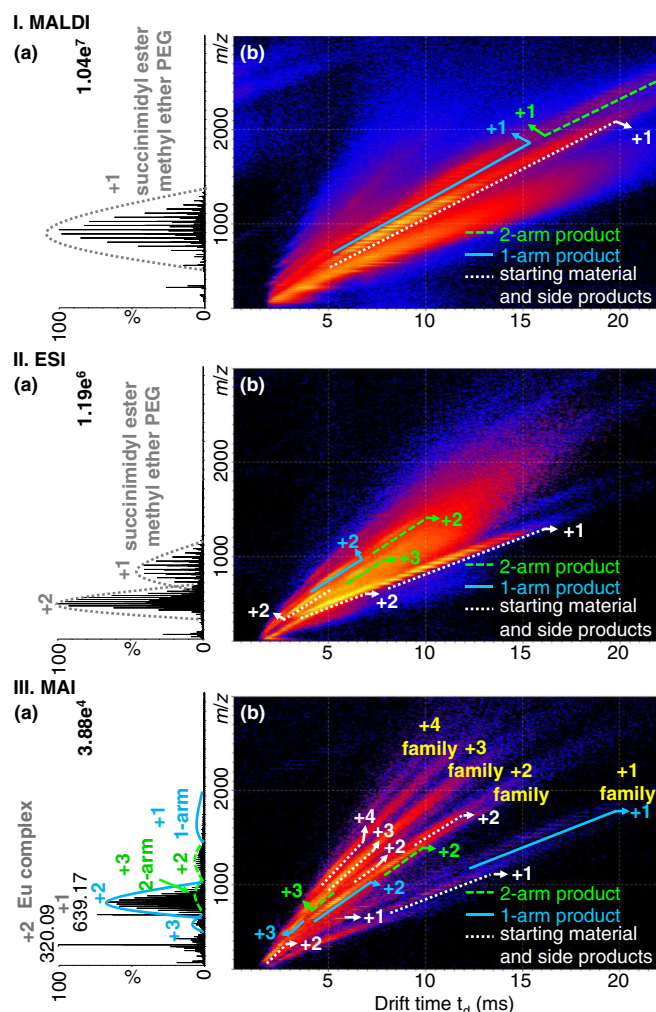


Figure 1. (a) Mass spectra and (b) 2-dimensional plots of the crude Eu-PEG sample analyzed by IMS-MS using different ionization methods of (I) MALDI using the matrix DCTB, (II) ESI, and (III) MAI using the matrix 1,2-DCB. Polymer distributions are denoted as starting materials and side products (grey, dotted), 1-arm Eu-PEG complex (blue, solid), and 2-arm Eu-PEG complex (green, dashed). Relative ion abundance is indicated by the color intensity imbedded into the 2-dimensional plot, with blue being the lowest, red intermediate, and yellow the highest abundance. Data were obtained on the intermediate pressure MALDI source (MALDI) and the atmospheric pressure Z-Spray source (ESI and MAI) of the Waters SYNAPT G2. Inset regions of MAI-IMS-MS 2-D plots are provided in Figure 2 and Supplementary Figure S8

became possible after extraction from the 2-D plot (Supplementary Figure S5), allowing separation of isobars through the use of drift times. Identification of europium-containing species was assisted by the unique isotopic distribution of europium. Multiply charged ions were generated through both alkali cation adduction and protonation for all species observed, contributing to convoluted datasets.

MAI-IMS-MS of the crude sample using 1,2-DCB as a matrix provided abundant multiply charged ion distributions of protonated 2-arm and 1-arm Eu-PEG relative to the unreacted starting material that were directly observable in the total mass spectrum

(Figure 1.III.a). Neither 3-NBN nor bronopol performed well (Supplementary Figure S6) relative to the matrix 1,2-DCB. The chemical background in MAI-IMS-MS using the matrix 1,2-DCB is minute (Figure 1.III.b), especially relative to MALDI and ESI (Figure 1.I.b and II.b). Additionally, charge states observed are higher (+3 and +4) than those detected using ESI (primarily +2), resulting in cleanly separated distributions in the 2-D plot (Figure 1.III.b). Gas-phase separation of the oligomer distributions in the t_d and m/z dimension allowed extraction of the respective mass spectra from the charge state families shown in Figure 1.III.b.

An expanded view of a region of Figure 1.III.b where ions overlap in the m/z dimension is shown in Figure 2a. The extraction of the drift time information of the individual oligomers in Figure 2b demonstrates that these isobars have a drift time difference of 0.77 ms and are cleanly baseline-resolved. Displaying the m/z dimension of the individual oligomers from the 2-D plot reveals two separate isotopic distributions (Figure 3a) represented by the blue squares and green asterisks, each containing the unique isotopic distribution inherent to europium. The extracted mass spectra can be represented as ‘nested’ datasets in units $t_d(m/z)$, where t_d is the drift time in milliseconds [54]. The two distinct europium-containing isotopic distributions corresponding to the 2-arm (Figure 3b.1) and 1-arm (Figure 3b.2) Eu-PEG conjugates were extracted, revealing isobars at

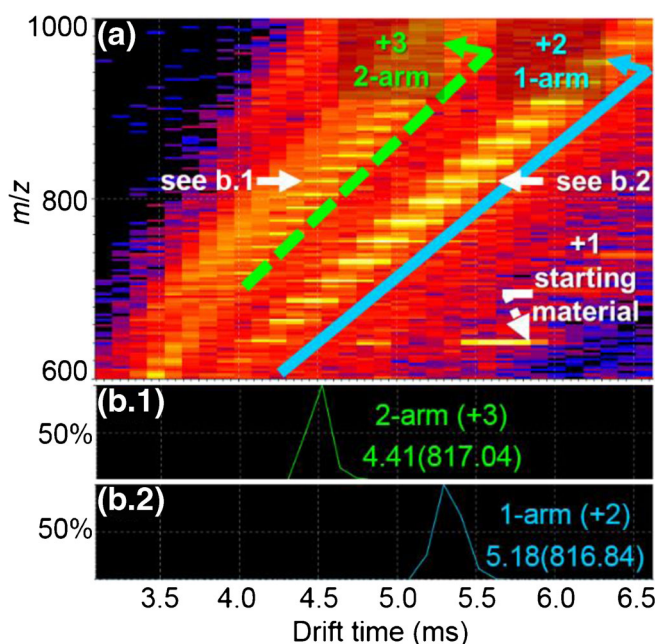


Figure 2. Inset region of the (a) 2-dimensional plot obtained by MAI-IMS-MS of the Eu-PEG crude sample in Figure 1.III. (b) Extracted drift times of (b.1) 4.41 ms and (b.2) 5.18 ms corresponding to the oligomer ions of the (b.1) 2-arm and (b.2) 1-arm products with a common ion at a m/z of 817. Polymer distributions are denoted as 1-arm Eu-PEG complex (blue, solid) and the 2-arm Eu-PEG complex (green, dashed). Relative ion abundance is indicated by the color intensity imbedded into the 2-dimensional plot, with blue being the lowest, red intermediate, and yellow the highest abundance

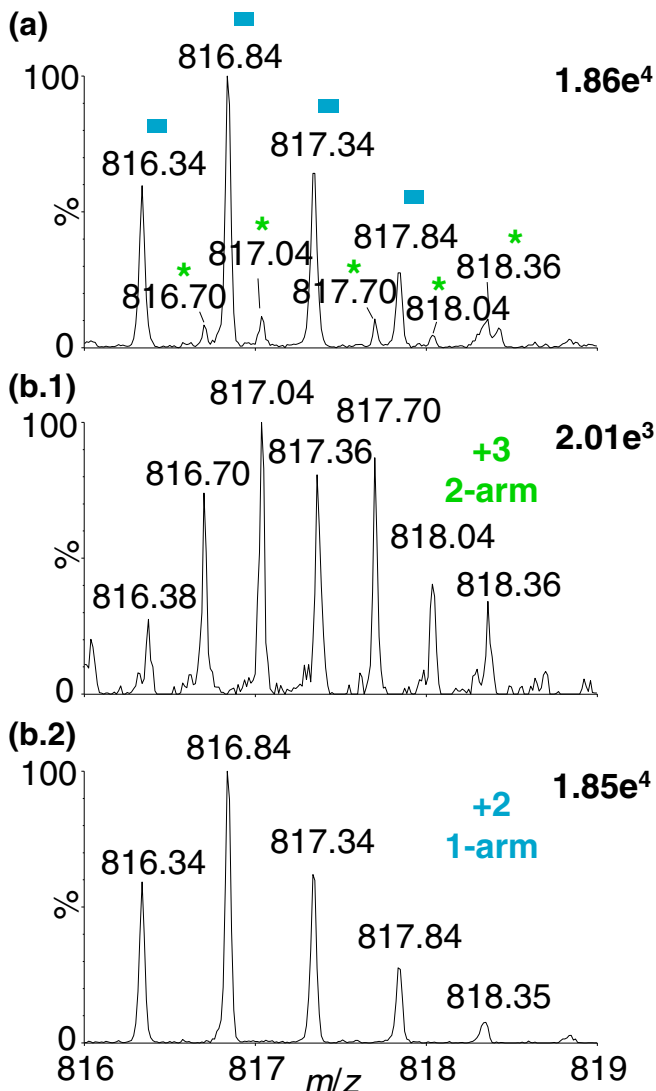


Figure 3. Mass spectra of isotopic distributions for extracted regions from Figure 2a of (a) the Eu-PEG crude sample, (b.1) extracted 2-arm (+3 charge state), and (b.2) extracted 1-arm (+2 charge state) species. The blue squares and green asterisks in (a) denote ions belonging to the isotopic distributions of the 1-arm and 2-arm Eu-PEG complexes in (b) and (c), respectively. Ion abundance of the highest ion peak is indicated at the top right corner

4.41(817.0) for the triply charged 2-arm species and 5.18(816.8) for the doubly charged 1-arm species. A linear trend between t_d and m/z is observed between oligomers of the doubly and triply charged families (Supplementary Figure S7). The plot of t_d versus m/z for the 2- and 1-arm Eu-PEG doubly charged products was fit using a linear regression and extended, suggesting a potential similarity between the shape of 2- and 1-arm Eu-PEG products with similar size and charge.

Interestingly, an additional oligomer distribution with europium-containing isotopic distributions was observed using MAI-IMS-MS (Figure 1.III.a, top left) but not with MALDI- or ESI-IMS-MS. The inset region (Supplementary Figure S8) shows low abundant distributions of triply (e.g. m/z 1028, Figure 4e.1)

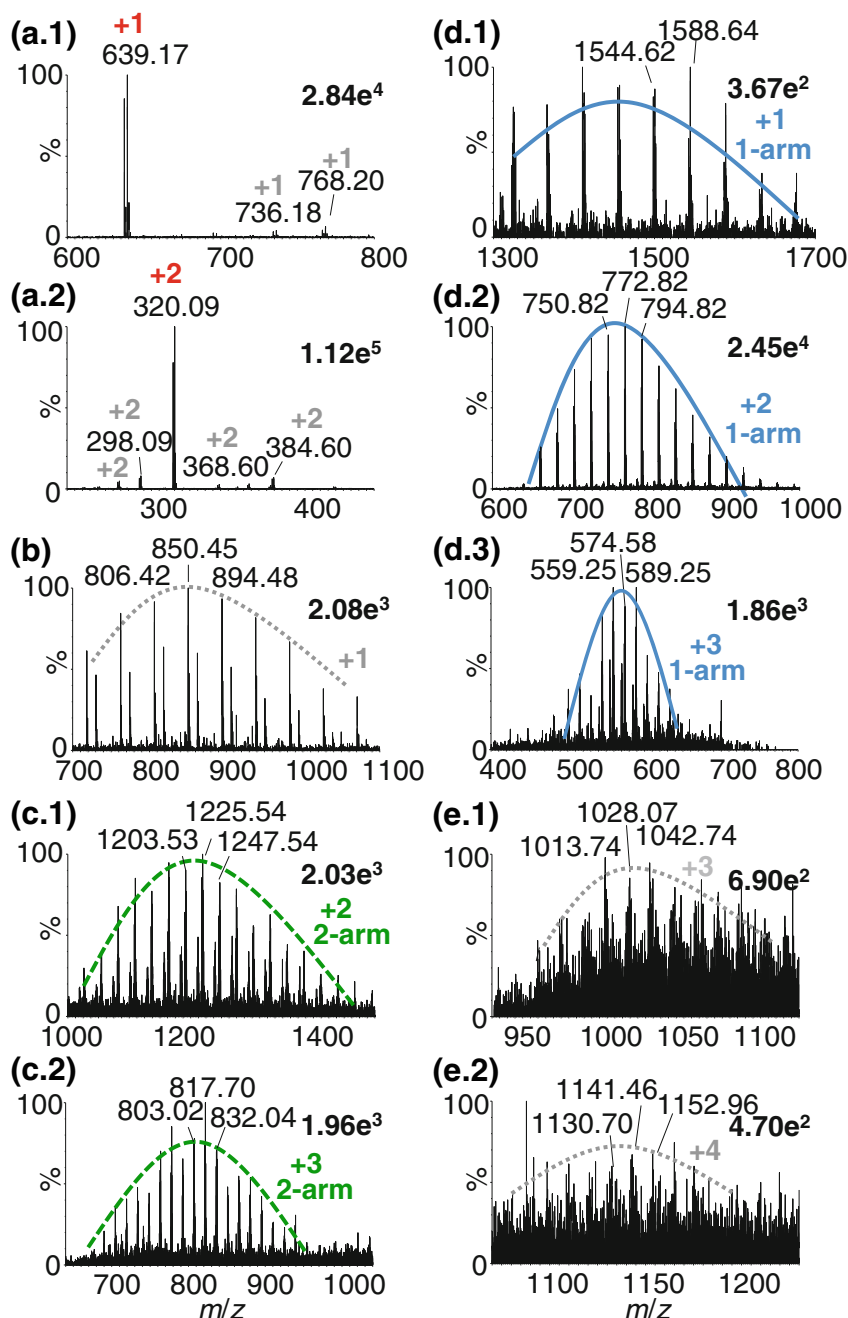


Figure 4. Extracted MAL mass spectra from Figure 1.III.b: (a) europium starting material, (b) PEG starting material, (c) 2-arm product, (d) 1-arm product, and (e) PEG starting material with respective charge states. Polymer distributions are denoted as: PEG starting material (grey, dotted), 1-arm Eu-PEG complex (blue, solid), and the 2-arm Eu-PEG complex (green, dashed). Ion abundance of the highest ion signal is indicated at the top right corner. Data were obtained on the atmospheric pressure Z-Spray source of a Waters SYNAPT G2

and quadruply (e.g., m/z 1141, Figure 4e.2) charged ion families containing PEG. The m/z values calculate to molecular weight distributions centered around 3.0 and 4.4 kDa, suggesting 3- and 4-arm Eu-PEG species exist, possibly a consequence of unreacted 1,4,7,10-tetraazacyclododecane starting material from early in the synthesis [53]. This starting material has up to four reactive sites for arm conjugation. Examples of the extracted mass spectra are provided for different charge state distributions of the intentional and unintentional products and the

individual starting materials, including the singly and doubly charged ions of the europium starting material (Figure 4a.1-2); singly charged PEG starting material (Figure 4b); doubly and triply charged 2-arm Eu-PEG product (Figure 4c.1-2); and singly, doubly, and triply charged 1-arm Eu-PEG product are shown (Figure 4d.1-3). Contrary to MALI, the low ion abundance of the 2- and 1-arm Eu-PEG product relative to the PEG starting material observed using MALDI and especially ESI, coupled with an inability to purify the sample, prompted a study to obtain an

estimate of the relative quantity of europium species in the crude sample.

Verification of Selective Ionization of Europium-Containing Oligomer Compounds in Crude Sample

A quantitative ICP-MS experiment was performed to estimate the mass percentage by weight of europium in the crude sample. ICP-MS revealed 0.8% of the crude sample by mass was europium (Supplementary Table S1 and Supplementary Figure S9). This value accounts only for the mass of europium, not for the mass of europium *and* the ligand. By assuming that all of the europium is bound within only the 2- or 1-arm ligand, a maximum mass percentage of Eu-PEG species was estimated. Adjusting the mass percent of europium for the mass of the 2- or 1-arm Eu-PEG complex, the mass percentage of Eu-PEG species present can be estimated to be a maximum of 10% assuming all europium is chelated within the 2-arm species and a maximum of 6% assuming all of the europium is chelated within the 1-arm species. Clearly, the majority of crude sample is not the Eu-PEG products. The initial experiments of the crude sample described above were performed on two different SYNAPT G2 instruments using MALDI- relative to ESI- and MAI-IMS-MS on different days. This prompted a study employing the three different ionization methods consecutively on the same mass spectrometer using the same sample that was used in the ICP-MS study.

The MALDI mass spectra showed a dependence on the relative laser fluence applied to the matrix:analyte sample initiating the ionization process (Supplementary Figure S10.I.a and Supplementary Figure S11) affording relative ratios of ion intensities of the Eu-PEG products and the PEG starting material. At low laser fluence, the ion intensity of the 2-arm Eu-PEG product was ~40% as intense as the PEG starting materials (Supplementary Figure S11.a). With increasing laser fluence, the PEG starting material dominated the mass spectrum, reducing the relative signal intensity of Eu-PEG to ~10% of the intensity of the PEG starting material signal (Supplementary Figure S11.c).

The ESI mass spectra showed the expected protonated, potassiated, and sodiated PEG starting materials, both singly and doubly charged, in high ion abundance (Supplementary Figure S10.II.a). Using the t_d dimension (Supplementary Figure S10.II.b) also made the protonated, potassiated, and sodiated doubly charged 2- and 1-arm Eu-PEG ions observable despite the intense chemical background. The Eu-PEG product signal intensity is ~5% as the intensity of the PEG starting material signal.

The MAI mass spectra revealed the europium containing products in high ion abundance, detected as protonated 2-arm Eu-PEG doubly and triply charged and the 1-arm species triply charged (Supplementary Figure S10.III.a). Extraction of the mass spectra from the 2-D plot also identified both doubly charged PEG starting materials and side products in low abundance (Supplementary Figure S10.III.b). The multiply charged ions obtained using MAI were through protonation and not metal cation

adduction. The intensity of the Eu-PEG signal is ~800% as abundant as the PEG intensity of the starting material signal.

Discussion

Ionization

Considering that the ICP-MS results estimated a maximum of 10% of the sample is Eu-PEG and that other europium-based species were detected only with MAI, it is obvious that the MAI matrix 1,2-DCB selectively ionized europium-containing complexes from the crude sample relative to starting materials that do not contain europium. The difference in results using three ionization methods is intriguing: the ion abundances of Eu-PEG relative to the PEG starting material are ~5% ESI, ~10%–40% MALDI, and ~800% MAI. An interesting point is that when *lower* energy was imparted using MALDI at lower laser fluence, the relative ratios of the ion intensities between Eu-PEG products and the PEG starting material are overrepresented. The overrepresentation with MALDI is similar, but not as drastic as what was observed when using MAI, in which no laser is used. A commonality between MALDI and MAI is the use of a solid matrix, which may attribute to the mutual overrepresentation.

Eu-PEG species were detected in MAI and MALDI, but when ESI was used, Eu-PEG species were only detectable in low abundance relative to other species. It is difficult to attribute results to the ion suppression issue described in MALDI and ESI [55, 56] considering the ICP-MS results, showing that MAI and MALDI both overrepresented the concentration of Eu-PEG products. Additionally, initial results from ionizing thermometer molecules using MAI indicate that MAI is harsher for small molecules than ESI [57] and the fact that typical capillary and extraction voltages were applied making fragmentation of the Eu-PEG product during ionization unlikely because of the inherently softer ionization afforded by ESI relative to MALDI [58].

The charge separation process in MAI is hypothesized to be a sublimation-driven process through crystal fracturing of compounds that triboluminescence [1,2,10]. Some europium-containing compounds are known to triboluminesce [59–63], potentially contributing to the overrepresentation of the europium-containing product. However, further experimentation is needed to support whether or not europium's triboluminescent properties contribute to the overrepresentation observed. The increase in ion abundance of europium-containing complexes using 1,2-DCB as a MAI matrix relative to other MAI matrices suggests that certain matrices enhance the specificity for ionizing different materials. Specificity for certain matrices was reported in the development of MALDI (e.g., sinapinic acid for high mass proteins [64], DCTB for fullerenes and polymers [65], and tetracyanoquinodimethane for polycyclic aromatic hydrocarbons [66]), and ESI (e.g., metal salts [67] and peptides [4]). This is similar to questions raised for ESI and MALDI here and in other work [4, 56, 68–71]. An explanation of how MAI can selectively ionize europium-containing species is of interest and may be determined with future work.

IMS Separation

The efficient ionization using the MAI matrix 1,2-DCB provided analytically useful IMS-MS results and allowed the analyses of the 2-arm complex separated from the 1-arm complex for the first time. The unsuccessful separation of the 2-arm product from the 1-arm product when using SEC suggests that the 1-arm and 2-arm species are of similar shape. Comparison of the drift times of the 1- and 2-arm Eu-PEG product directly is impossible because there are no regions where they both overlap in m/z . The extrapolated data from the linear trend observed between the t_d and m/z of oligomers (Supplementary Figure S7) predicts that when mass and charge are similar, the 2- and 1-arm Eu-PEG products are of closely related shape (Supplementary Figure S12). This is interesting because the *trans*-addition of the PEG arm does not significantly alter the shape of the europium complex. The 2- and 1-arm Eu-PEG products having closely related shapes provides an explanation for difficulties in purification using SEC, highlighting the utility of using three-dimensional separation by charge, size, and shape inherent to IMS [54, 72].

Conclusions

Gas-phase separation methods were used to examine a crude Eu-PEG sample, where conventional liquid-phase separation and ionization methods for MS had difficulties or failed. The use of proper matrix material to ionize by MAI coupled with IMS-MS provides rapid, information-rich datasets with the ability to be “extracted” for analysis of complex polymers. The use of IMS-MS provided evidence that the proposed 2-arm Eu-PEG complex was synthesized. Because of the formation of the multiply charged ions, MAI-IMS-MS is a useful tool in the analyses of complex polymeric systems that differ by slight changes in structure, such as 2- and 1-arm conjugates. Additional information was obtained from mass spectra generated by MAI, but not MALDI or ESI, including the presence of PEG-containing side products and Eu-PEG conjugates containing additional PEG arms. Future work is directed towards understanding the selective and potentially soft nature of MAI for nonvolatile molecules and the unexpected differences in the formation of multiply charged ions from the solution state (ESI) and the solid state (MAI) conditions.

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